

Subject Description Form

Subject Code	COMP3421
Subject Title	Web Application Design and Development
Credit Value	3
Level	3
Pre-requisite / Co-requisite / Exclusion	Pre-requisite: COMP1011/COMP1012/ENG2002
Objectives	The objectives of this subject are to: 1. introduce the principles, architecture, and impact of modern web applications in enabling distributed, accessible, and interactive computing environments; 2. equip students with the skills to analyse, design, and implement responsive, secure, and data-driven web applications using contemporary front-end and back-end technologies; and 3. review state-of-the-art web development practices, including cloud-based deployment, real-time communication, and emerging trends in web and mobile integration.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: <i>Professional / Academic Knowledge and Skills</i> (a) differentiate the key components and roles of front-end, back-end, and database systems in distributed web application architecture; (b) apply the concepts of web services, APIs, and related technologies in modern web development tasks; (c) demonstrate proficiency in client-side and server-side programming using contemporary tools and frameworks (e.g., JavaScript/TypeScript, React, Node.js); (d) design, develop, and deploy innovative, responsive, and interactive web applications; (e) evaluate and apply emerging web technologies, including cloud-based deployment, real-time communication, and mobile integration; (f) understand current and future trends in web technology, including progressive web applications, AI integration, and cross-platform development; <i>Attributes for All-roundedness</i> (g) communicate effectively through project presentations, system demonstrations, and technical reports; (h) demonstrate independent learning skills and apply newly acquired knowledge to solve non-routine technical problems; (i) accept responsibility and accountability for achieving personal and group outcomes while exhibiting leadership in a project team; and (j) apply critical thinking and creativity in integrating diverse computing technologies into innovative web applications.

Subject Synopsis/ Indicative Syllabus	Topic
	1. Introduction to Distributed Web Application Architecture This topic explores the evolution of client–server computing and its role in shaping modern Internet applications. It covers the main components of web infrastructure, such as web servers, application servers, database servers, and API gateways, and examines the differences between two-tier, three-tier, and multi-tier architectures. The underlying network protocols and infrastructure supporting distributed web applications are also discussed.
	2. Web-Based Application Development This topic introduces the web as a platform for distributed computing, explaining the web application model, protocols, and hypertext technologies. It examines the HTTP request–response cycle, data representation formats, and the development of interactive applications using modern client-side and server-side technologies, including HTML5, CSS3, JavaScript/TypeScript, and contemporary frameworks for back-end development.
	3. Interactive and Data-Driven Web Applications This topic focuses on techniques for creating responsive, data-driven experiences. It covers asynchronous communication methods such as AJAX, Fetch API, and WebSockets, as well as approaches to building single-page and progressive web applications. The integration of databases and APIs for dynamic content delivery is also addressed.
	4. Latest and Future Web Computing This topic reviews recent advancements in web technologies, including cloud-based deployment, mobile and cross-platform integration, and real-time application development. It also discusses emerging trends such as AI-enhanced web applications, Web 3.0 concepts, and the evolving role of intelligent and decentralized Internet computing.
Teaching/ Learning Methodology	This subject emphasises the design and technical aspects of web application development. It is intended to equip the student with knowledge and practical experience on how to complete a web-based application. The lectures will be used to deliver course material that will be practised/reinforced during the labs and tutorials.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed										
			a	b	c	d	e	f	g	h	i	j	
	Continuous Assessment	30%											
	1. Assignments, Tests & Projects		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
	Examination	70%	✓	✓				✓		✓			
	Total	100 %											
Student Study Effort Expected	Class contact:												
	▪ Lectures								26 Hrs.				
	▪ Tutorials/Lab								13 Hrs.				
	Other student study effort:												
	▪ Assignments, Tests, Projects, Exams								80 Hrs.				
	Total student study effort								119 Hrs.				
Reading List and References	Reference Books:												
	[1].Robbins, Jennifer Niederst, <i>Learning Web Design: A Beginner’s Guide to HTML, CSS, JavaScript, and Web Graphics</i> , 5th Edition, O’Reilly Media, 2018.												
	[2].Haverbeke, Marijn, <i>Eloquent JavaScript: A Modern Introduction to Programming</i> , 4rd Edition, No Starch Press, 2024.												
	[3].Nixon, Robin, <i>Learning PHP, MySQL & JavaScript</i> , 6th Edition, O’Reilly Media, 2021.												
	[4].Flanagan, David, <i>JavaScript: The Definitive Guide</i> , 7th Edition, O’Reilly Media, 2020.												
	[5].Kurose, James F., and Keith W. Ross, <i>Computer Networking: A Top-Down Approach</i> , 8th Edition, Pearson, 2020.												